LAB WORKSHEET 02

CTEC 22043 – Object-Oriented Programming

CT/2022/064 - GUNARATHNE K.A.T.W.W.

Q01.

.java

```java
import java.util.Scanner;

public class Q1 {
    static void main(String[] args) {
        double A, B, C, X, Y, R;
        final double PI = 3.14;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("\nEnter a value for 'A': ");
        A = input.nextDouble();

        System.out.print("Enter a value for 'B': ");
        B = input.nextDouble();

        System.out.print("Enter a value for 'C': ");
        C = input.nextDouble();

        System.out.print("Enter a value for 'X': ");
        X = input.nextDouble();

        System.out.print("Enter a value for 'Y': ");
        Y = input.nextDouble();

        System.out.print("Enter a value for 'R': ");
        R = input.nextDouble();

        //Part a
        double expression1 = Math.pow(B, 2) + (4 * A * C);
        double answer1 = Math.sqrt(expression1);
        System.out.println("\nThe square root of (B^2 + 4AC) " + answer1);

        //Part b
        double expression2 = X + (4 * Math.pow(Y, 3));
        double answer2 = Math.sqrt(expression2);
        System.out.println("The square root of (X + 4Y^3) is " + answer2);

        //Part c
        double expression3 = X * Y;
        double answer3 = Math.cbrt(expression3);
        System.out.println("The cube root of the product of X and Y is " + expression3);

        //Part d
        double area = PI * (R * R);
        System.out.println("Are of the circle is " + area);
    }
}
```

Q02.

```java
import java.util.Scanner;

public class Q2 {
    public static void main(String[] args) {
        System.out.println("\n--This program converts centimeter to feet and inches--");
        int cm, feet, inch;
        double inches;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the length in centimeter: ");
        cm = input.nextInt();

        //calculation
        inches = cm / 2.54;
        feet = (int) (inches / 12);
        inch = (int) (inches % 12);

        //print
        System.out.println(cm + " cm = " + feet + " feet " + inch + " inches");
    }
}
```

Q03.

```java
import java.util.Scanner;

public class Q3 {
    public static void main(String[] args) {
        System.out.println("\n--This program prints out the temperature in Fahrenheit--");
        double celsius, fahrenheit;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the temperature in Celsius: ");
        celsius = input.nextDouble();

        //calculation
        fahrenheit = (1.8 * celsius) + 32;

        //print
        System.out.println("Temperature is " + fahrenheit + "F.");
    }
}
```

Q04.

```java
import java.util.Scanner;

public class Q4 {
    public static void main(String[] args) {
        System.out.println("\n--This program calculates the number of calories you need in one day--");
        double weight, calories;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your body weight (pounds): ");
        weight = input.nextInt();

        //calculation
        calories = weight * 19;

        //print
        System.out.println("You need " + calories + "cal per day.");
    }
}
```

Q05.

```java
import java.util.Scanner;

public class Q5 {
    public static void main(String[] args) {
        System.out.println("\n--This program converts the temperature in Celsius--");
        double fahrenheit, celsius;

        //getting input
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the temperature in Fahrenheit: ");
        fahrenheit = input.nextInt();

        //calculation
        celsius = (((double) 5 / 9) * (fahrenheit - 32));

        //print
        System.out.println("Temperature is " + celsius + "C.");
    }
}
```

Q06.

```java
import java.util.Scanner;

public class Q6 {
    public static void main(String[] args) {
        System.out.println("\n--This program prints your age--");
        int bornYear, age;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your born year: ");
        bornYear = input.nextInt();

        //calculation
        age = 2026 - bornYear;

        //print
        System.out.print("You were born in " + bornYear + " and you will be (are) " + age + " this year.");
    }
}
```

Q07.

```java
import java.util.Scanner;

public class Q7 {
    public static void main(String[] args) {
        System.out.println("\n--This program prints your Body Mass Index (BMI)--");
        int h, w;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your height (cm): ");
        h = input.nextInt();

        System.out.print("Enter your weight (kg): ");
        w = input.nextInt();

        //calculation
        double bmi = w / (Math.pow ((h / 100.0), 2));
        System.out.println(bmi);
        String status;
        if (bmi >= 20 && bmi <= 25) {
            status = "is normal.";
        } else {
            status = "should be within 20 and 25.";
        }

        //print
        System.out.println("Your BMI is " + bmi + " and it " + status);
    }
}
```

Q08.

```java
import java.util.Scanner;

public class Q8 {
    public static void main(String[] args) {
        System.out.println("--\nThis program computes the volume of a sphere--");
        double r, v;

        //getting inputs
        final double PI = 3.14;
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the radius of sphere: ");
        r = input.nextDouble();

        //calculation
        v = (4.0 / 3) * (PI / (Math.pow(r, 3)));

        //print
        System.out.println("Volume of the sphere is " + v);
    }
}
```

Q09.

```java
import java.util.Scanner;

public class Q9 {
    public static void main(String[] args) {
        System.out.println("\n--This program computes the amount of money from an investment--");
        double money, years, rate, totalMoney;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Hou much money do you want to invest: ");
        money = input.nextDouble();

        System.out.print("For how many years: ");
        years = input.nextDouble();

        System.out.print("What is the rate (%): ");
        rate = input.nextDouble();

        //calculation
        totalMoney = money * Math.pow((1 + rate / 100.0), years);

        //print
        System.out.print("The amount of money you can earn : Rs. " + totalMoney);
    }
}
```

Q10.

```java
import java.util.Scanner;

public class Q10 {
    public static void main(String[] args) {
        System.out.println("\nThis program is a loan calculator");

        double loanAmount, annualInterestRate, monthlyInterestRate, monthlyPayment, totalPayment;
        int loanPeriod, numberOfPayments;
        final int MONTHS_IN_YEAR = 12;

        //getting inputs
        Scanner input = new Scanner(System.in);

        System.out.print("Loan amount: ");
        loanAmount = input.nextDouble();

        System.out.print("Annual interest rate (%): ");
        annualInterestRate = input.nextDouble();

        System.out.print("Loan period (years): ");
        loanPeriod = input.nextInt();

        //calculation
        monthlyInterestRate = annualInterestRate / 100.0 / MONTHS_IN_YEAR;

        numberOfPayments = loanPeriod * MONTHS_IN_YEAR;

        monthlyPayment = (loanAmount * monthlyInterestRate) / (1 - Math.pow(1 / (1 + monthlyInterestRate), numberOfPayments));

        totalPayment = monthlyPayment * numberOfPayments;

        //print
        System.out.println("Your monthly payment is Rs. " + monthlyPayment);
        System.out.print("Your total payment is Rs. " + totalPayment);
    }
}
```